

R starter sheet for descriptive analysis

If you are ready to move beyond Excel — every command you need for Day 2 and Day 3.

You do not need R to complete this course. But R gives you something Excel cannot: a **script**, which is a permanent, re-runnable record of every decision you made. When a reviewer asks what you did, you send them the file.

Getting started

```
# Install once, then load every session
install.packages(c("tidyverse", "janitor"))
library(tidyverse)
library(janitor)

# Read your data (put the CSV in your working folder)
tb <- read_csv("tb_cohort_timor_leste_2024_RAW.csv")

# First look - ALWAYS do these three
glimpse(tb)      # what columns, what types
summary(tb)      # min, max, median, missing
nrow(tb)         # how many rows - does it match your register?
```

Cleaning — the same seven checks as Factsheet 4

```
# 1 & 2. Duplicates
sum(duplicated(tb$record_id))
tb <- distinct(tb, record_id, .keep_all = TRUE)

# 3. Every distinct value in a category column (with NA shown)
tb %>% count(sex)
tb <- tb %>% mutate(sex = case_when(
  sex %in% c("M", "Male", "male", "1") ~ "Male",
  sex %in% c("F", "Female", "female") ~ "Female",
  TRUE ~ NA_character_))

# 4. Impossible values
range(tb$age_years, na.rm = TRUE)
tb <- tb %>% mutate(age_years = if_else(age_years > 110, NA_real_, age_years))

# 6. Missing data per column
colSums(is.na(tb))
```

Describing

```
# Continuous: mean, median, spread, and how much is missing
tb %>% summarise(
  n      = sum(!is.na(distance_to_facility_km)),
  missing = sum(is.na(distance_to_facility_km)),
  mean   = mean(distance_to_facility_km, na.rm = TRUE),
  median = median(distance_to_facility_km, na.rm = TRUE),
  q1     = quantile(distance_to_facility_km, .25, na.rm = TRUE),
  q3     = quantile(distance_to_facility_km, .75, na.rm = TRUE))

# Categorical: counts AND percentages, with the denominator visible
tb %>% tabyl(outcome) %>% adorn_pct_formatting(digits = 1)
```

```
# Cross-tabulation with row percentages
tb %>% tabyl(sex, outcome) %>%
  adorn_percentages("row") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

Treatment success — and why the denominator is a line of code

```
# Be explicit about who you are counting. This is your denominator decision,
# written down permanently, where a reviewer can see it.
evaluated <- tb %>% filter(!is.na(outcome), outcome != "Not evaluated")

evaluated %>% summarise(
  successes = sum(outcome %in% c("Cured", "Treatment completed")),
  N          = n(),
  pct        = round(100 * successes / N, 1))
```

Plotting

```
# Histogram of a continuous variable
ggplot(tb, aes(x = distance_to_facility_km)) +
  geom_histogram(binwidth = 5, fill = "#0F7B7B", colour = "white") +
  labs(title = "Distance from home to treatment facility",
        subtitle = "TB patients, Timor-Leste, 2024 (n = 180)",
        x = "Distance (km)", y = "Number of patients") +
  theme_minimal()

# Bar chart of a category, sorted largest to smallest
tb %>% count(municipality) %>%
  ggplot(aes(x = reorder(municipality, n), y = n)) +
  geom_col(fill = "#1F3864") + coord_flip() +
  labs(x = NULL, y = "Number of patients registered") +
  theme_minimal()

# Box plot - shows spread, which a bar of means hides
ggplot(tb, aes(x = outcome, y = distance_to_facility_km)) +
  geom_boxplot(fill = "#E4F2EF") + coord_flip() +
  labs(x = NULL, y = "Distance to facility (km)") +
  theme_minimal()

# Save at publication quality
ggsave("figure1.png", width = 18, height = 11, units = "cm", dpi = 300)
```

TIP

na.rm = TRUE tells R to ignore missing values. That is convenient, but it also silently changes your denominator. Always run `colSums(is.na(tb))` first so you know exactly how many records you are dropping, and report that number.